

CNN-GP-Supplementary Material

I. CNN MODULE ARCHITECTURE ANALYSIS

The CNN module is designed as a feature-oriented representation learner for high-dimensional financial time-series data. The original data can be represented as $X:T \times N \times D$ where T , N , and D denote the number of trading dates, stocks, and original features, respectively. Instead of directly performing temporal convolution across consecutive trading days, our CNN processes the multi-feature representation of each stock at each trading date and transforms the original D -dimensional feature space into a compact M -dimensional latent representation, producing $Z:T \times N \times M$.

As illustrated in Fig. S1, the feature extractor consists of four consecutive one-dimensional convolutional blocks. The channel dimensions are progressively expanded from 1 to 64, 64 to 128, 128 to 128, and 128 to 256, respectively. All convolutional layers use a kernel size of 1. Each convolutional layer is followed by Batch Normalization, ReLU activation, Max Pooling, and Dropout. After the four convolutional blocks, Global Average Pooling aggregates the high-level feature responses to obtain a compact 256-dimensional representation. An additional 1×1 Conv1D projection layer then maps this representation from 256 channels to the M -dimensional latent feature space, which serves as the input to the subsequent GP module. A classification layer is additionally employed to categorize stocks into three coarse-grained groups: profitable, risky, and lossy. A key characteristic of this architecture is the use of kernel-size-1 convolution along the feature representation rather than temporal convolution over neighboring dates. A kernel size of 1 does not explicitly combine adjacent feature positions within the convolution operation. Instead, it performs a learnable channel transformation at each feature position. As the network becomes deeper, the channel dimension is progressively expanded, enabling increasingly rich nonlinear transformations of the original stock characteristics. The subsequent pooling operations progressively aggregate and compress these transformed feature responses, while Global Average Pooling produces a compact global representation.

This design is intentional because the original financial variables are heterogeneous indicators rather than naturally ordered spatial or temporal neighbors. For example, price-related, volume-related, volatility-related, and momentum-related variables may occupy adjacent positions in the input vector, but their adjacency does not necessarily imply the same local relationship as neighboring pixels in an image or neighboring observations in a temporal sequence. Therefore, directly applying a larger convolutional kernel across the feature vector would implicitly impose a local-neighborhood assumption on the ordering of heterogeneous financial variables. The kernel-size-1 convolutions themselves do not impose learned local interactions

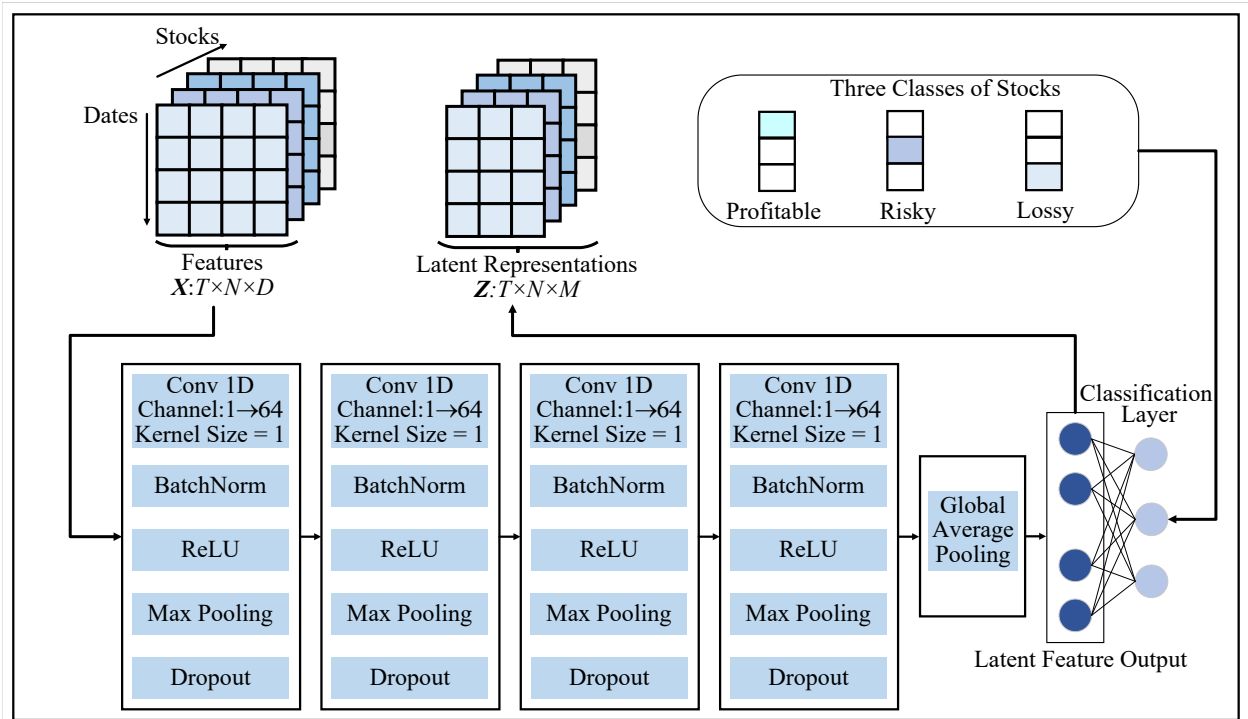


Fig. S1. CNN Module Architecture for Latent Feature Extraction.

between adjacent feature positions. Instead, they perform pointwise channel transformations, while the subsequent pooling layers progressively aggregate and compress the transformed responses.

This point is particularly important for understanding how our CNN differs from conventional CNN-based time-series models. Traditional temporal CNNs typically slide convolution kernels across consecutive timestamps to capture short-term temporal dependencies. In contrast, our CNN does not aim to explicitly model local temporal transitions inside the convolutional module. Instead, we treat each trading date as a time-indexed observation and focus on learning a compact representation of the high-dimensional stock state observed at that date. The overall prediction problem remains time-indexed through the chronological feature–target alignment, while the CNN itself operates on the cross-feature representation of each stock at each trading date and does not explicitly model temporal dependencies across consecutive dates.

Consequently, our approach represents a different way of applying CNNs to financial time-series analysis. Rather than assuming that temporal convolution is always the appropriate mechanism for extracting useful information from time-series data, CNN-GP separates the problem into two complementary stages. First, the CNN transforms the high-dimensional feature state at each timestamp into a compact and task-oriented latent representation. Second, GP performs symbolic alpha discovery over these learned representations. In this sense, the CNN serves as a representation-space constructor for the subsequent symbolic search rather than simply as a conventional end-to-end temporal predictor.

The coarse-grained three-class classification objective further plays an important role in shaping this latent space. The CNN is not trained merely to reconstruct or compress the original features. Instead, its latent representations are optimized to distinguish stocks according to their forward-return characteristics. Therefore, the learned M -dimensional representation is task-oriented: information useful for distinguishing profitable, risky, and lossy stocks is emphasized, whereas redundant or less relevant information in the original high-dimensional space can be suppressed. The latent features consequently provide GP with a substantially smaller and more discriminative search space than the original raw features.

This architecture also establishes a natural connection between representation learning and symbolic alpha mining. The CNN addresses the difficulty of searching directly over a large number of noisy and redundant financial variables, while GP converts the resulting latent representations into explicit symbolic expressions. Thus, the two components play complementary roles: the CNN performs feature abstraction and dimensionality reduction, whereas GP performs fine-grained symbolic relationship discovery. The resulting framework combines the expressive representation capability of neural networks with the structural transparency of symbolic models.

TABLE SI
STATISTICAL RESULTS OF CNN-GP AND COMPARATIVE METHODS

Dataset / Metric	Quality	CNN-GP	AlphaForge	D-Va	QFR	RiskMiner	SFAG
CSI500 RankIC	Mean	0.053020	0.043300	0.034100	0.038000	0.021400	0.018800
	Std	0.002953	0.003305	0.003155	0.006678	0.004364	0.004195
	Median	0.052650	0.043600	0.034500	0.037800	0.021350	0.017050
	p-value	—	2.46E-04+	1.83E-04+	2.46E-04+	1.83E-04+	1.83E-04+
CSI500 ALER	Mean	0.164800	0.089200	0.062200	0.038010	0.071200	0.011000
	Std	0.005587	0.002858	0.003055	0.005625	0.003078	0.000606
	Median	0.167200	0.089250	0.062300	0.038500	0.071050	0.011100
	p-value	—	1.82E-04+	1.82E-04+	1.82E-04+	1.82E-04+	1.80E-04+
CSI1000 RankIC	Mean	0.064600	0.052300	0.043500	0.044300	0.033700	0.034600
	Std	0.001678	0.002877	0.003337	0.002677	0.003349	0.001478
	Median	0.063950	0.052350	0.043500	0.044200	0.033500	0.034950
	p-value	—	1.83E-04+	1.83E-04+	1.83E-04+	1.83E-04+	1.83E-04+
CSI1000 ALER	Mean	0.157810	0.063200	0.064620	0.061800	0.053000	0.082600
	Std	0.002954	0.002378	0.002726	0.003036	0.002997	0.002980
	Median	0.157450	0.063600	0.065050	0.061850	0.053050	0.083150
	p-value	—	1.82E-04+	1.82E-04+	1.82E-04+	1.82E-04+	1.82E-04+
Friedman Rank		1.00	2.50	3.75	4.00	5.00	4.75

Note: Mean, standard deviation (Std), and median are calculated over 10 independent runs. The p -values are obtained using the two-sided Wilcoxon rank-sum test between CNN-GP and each competing method. “+”, “=”, and “—” indicate that CNN-GP performs significantly better than, shows no significant difference from, or performs significantly worse than the corresponding method, respectively, at the $p < 0.05$ significance level. The last row reports the Friedman mean rank, where a smaller rank indicates better overall performance.

TABLE SII
STATISTICAL RESULTS OF CNN-GP AND ABLATION METHODS

Dataset / Metric	Quality	CNN-GP	CNN-XGBoost	CNN	GP	XGBoost
CSI500 RankIC	Mean	0.053020	0.045630	0.044400	0.042800	0.032810
	Std	0.002953	0.002443	0.006196	0.002680	0.003199
	Median	0.052650	0.045650	0.042200	0.043200	0.032850
	p-value	—	4.40E-04+	9.11E-03+	1.83E-04+	1.83E-04+
CSI500 ALER	Mean	0.164800	0.114100	0.105300	0.108100	0.019330
	Std	0.005587	0.002937	0.002985	0.002072	0.002536
	Median	0.167200	0.114150	0.105100	0.108050	0.019150
	p-value	—	1.82E-04+	1.82E-04+	1.82E-04+	1.82E-04+
CSI1000 RankIC	Mean	0.064600	0.059200	0.058400	0.051200	0.049100
	Std	0.001678	0.002851	0.001534	0.003222	0.002928
	Median	0.063950	0.059200	0.058700	0.051750	0.049750
	p-value	—	1.83E-04+	1.83E-04+	1.83E-04+	1.83E-04+
CSI1000 ALER	Mean	0.157810	0.098480	0.129000	0.077200	0.054040
	Std	0.002954	0.002795	0.000459	0.002848	0.002422
	Median	0.157450	0.098300	0.129000	0.077350	0.053450
	p-value	—	1.82E-04+	1.80E-04+	1.82E-04+	1.82E-04+
Friedman Rank		1.00	2.25	3.00	3.75	5.00

Note: Mean, standard deviation (Std), and median are calculated over 10 independent runs. The p -values are obtained using the two-sided Wilcoxon rank-sum test between CNN-GP and each ablation method. “+”, “=”, and “−” indicate that CNN-GP performs significantly better than, shows no significant difference from, or performs significantly worse than the corresponding method, respectively, at the $p < 0.05$ significance level. The last row reports the Friedman mean rank, where a smaller rank indicates better overall performance.

II. STATISTICAL SIGNIFICANCE ANALYSIS

To further examine whether the performance improvements of CNN-GP are statistically reliable rather than caused by randomness across individual runs, we conduct statistical significance analysis based on 10 independent runs for each method. The mean, standard deviation, and median are reported to characterize the distribution of the experimental results.

For pairwise statistical comparisons, we employ a two-sided Wilcoxon rank-sum test between CNN-GP and each competing method. The symbols “+”, “=”, and “−” indicate that CNN-GP performs significantly better than, shows no significant difference from, or performs significantly worse than the corresponding method, respectively, at the $p < 0.05$ significance level. We also report the Friedman mean rank to evaluate the overall relative performance of all methods.

As shown in Table SI, CNN-GP consistently outperforms all five comparative methods on both CSI500 and CSI1000. For CSI500 RankIC, the Wilcoxon rank-sum p -values against AlphaForge, D-Va, QFR, RiskMiner, and SFAG are 2.46×10^{-4} , 1.83×10^{-4} , 2.46×10^{-4} , 1.83×10^{-4} , and 1.83×10^{-4} , respectively. For CSI500 ALER, all comparisons are also significant, with p -values ranging from 1.80×10^{-4} to 1.82×10^{-4} . The same conclusion holds on CSI1000, where all RankIC and ALER comparisons remain statistically significant. Moreover, CNN-GP obtains the best Friedman mean rank of 1.00, followed by AlphaForge (2.50), D-Va (3.75), QFR (4.00), SFAG (4.75), and RiskMiner (5.00).

Table SII reports the corresponding results for the ablation study. CNN-GP significantly outperforms CNN-XGBoost, CNN, GP, and XGBoost on both RankIC and ALER across CSI500 and CSI1000. Even the largest pairwise p -value, obtained for CNN-GP versus CNN on CSI500 RankIC ($p = 9.11 \times 10^{-3}$), remains clearly below the 0.05 significance threshold. All other pairwise p -values are below 4.40×10^{-4} . CNN-GP also obtains the best Friedman mean rank of 1.00, compared with 2.25 for CNN-XGBoost, 3.00 for CNN, 3.75 for GP, and 5.00 for XGBoost.

Overall, the Wilcoxon rank-sum tests and Friedman rankings consistently demonstrate that CNN-GP achieves statistically reliable and robust improvements over both the comparative methods and the ablation variants across the two datasets and two evaluation metrics. These results provide further evidence that the observed performance gains are systematic across independent runs rather than being driven by isolated favorable outcomes.

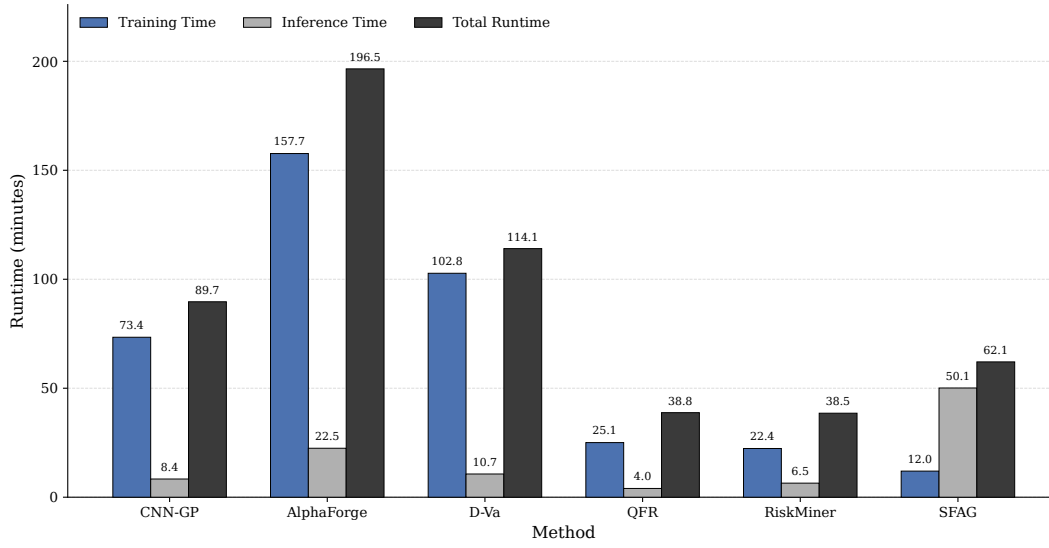


Fig. S2. Comparison of execution times between CNN-GP and the comparative methods.

III. RUNTIME ANALYSIS

To evaluate the practical computational cost of CNN-GP, we further compare the training time, inference time, and total runtime of the proposed method with both the competing methods and the ablation variants under the same experimental setting. The runtime comparison with the competing methods is shown in Fig. S2, while the comparison with the ablation variants is presented in Fig. S3. Note that the runtime is not separately reported for CSI500 and CSI1000, since factor values are generated for all stocks in a unified training and inference process; CSI500 and CSI1000 are separated only during the subsequent backtesting and performance evaluation.

As shown in Fig. S2, CNN-GP requires 89.7 minutes in total, compared with 196.5 minutes for AlphaForge, 114.1 minutes for D-Va, 38.8 minutes for QFR, 38.5 minutes for RiskMiner, and 62.1 minutes for SFAG. CNN-GP is therefore considerably faster than AlphaForge and D-Va, while QFR, RiskMiner, and SFAG require less total runtime. Although CNN-GP is not the fastest method overall, it consistently achieves higher RankIC and ALER on both CSI500 and CSI1000. These results indicate that CNN-GP provides a favorable trade-off between computational cost and predictive performance.

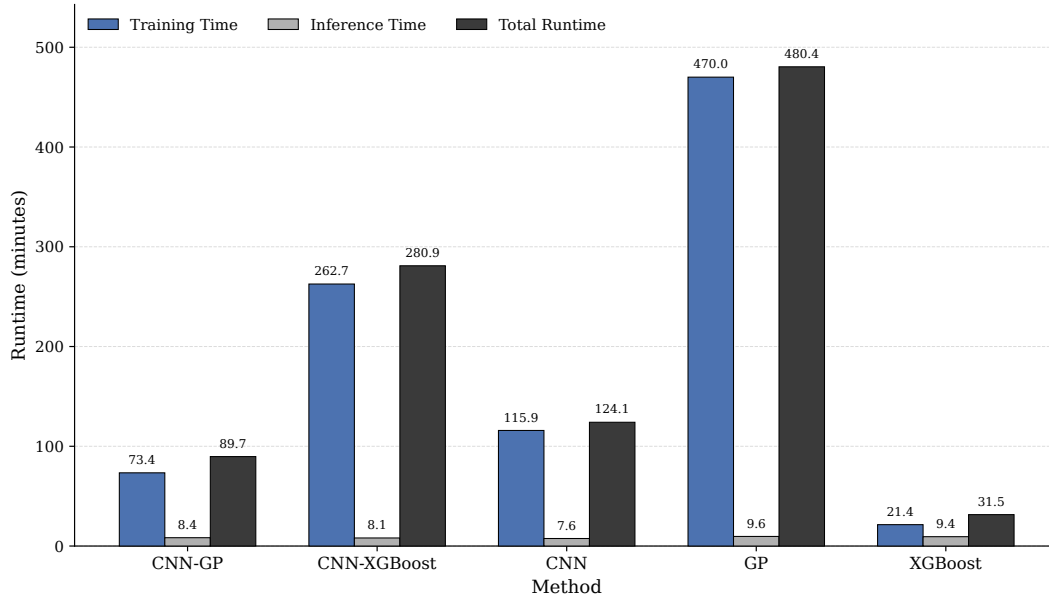


Fig. S3. Comparison of execution times between CNN-GP and the ablation variants.

As shown in Fig. S3, the ablation comparison further demonstrates the computational advantage of the proposed framework. CNN-XGBoost requires approximately 280.9 minutes in total, while the standalone CNN requires 124.1 minutes and GP requires 480.4 minutes. In comparison, CNN-GP completes the entire process in only 89.7 minutes. In particular, compared with directly applying GP to the original feature space, CNN-GP reduces the total runtime from 480.4 to 89.7 minutes, corresponding to an approximately 81.3% reduction in runtime. The total runtime is also approximately 27.7% lower than that of the standalone CNN and 68.1% lower than that of CNN-XGBoost. Together with the superior RankIC and ALER obtained by CNN-GP in the ablation experiments, these results indicate that the proposed integration improves not only predictive effectiveness but also computational efficiency.

Another observation is that the inference cost of CNN-GP remains relatively small. CNN-GP requires only 8.4 minutes for inference, which is comparable to or lower than several competing methods. This suggests that much of the computational effort is concentrated in the offline representation-learning and symbolic-search stages, whereas applying the learned model for subsequent evaluation does not introduce excessive inference overhead.

Overall, the runtime analysis shows that CNN-GP achieves a practical balance between computational cost and predictive performance. Compared with several competing methods, CNN-GP maintains moderate computational cost while achieving stronger RankIC and ALER performance. Compared with standalone GP and the ablation variants, the proposed framework is also substantially more computationally efficient. These results support the practical feasibility of combining compact representation learning with symbolic alpha mining.

IV. SENSITIVITY ANALYSIS OF THE LATENT FEATURE DIMENSION

The latent feature dimension determines the size of the representation passed from the CNN module to the subsequent GP-based symbolic search. In the original input space, each stock is represented by 52 financial features. The purpose of the CNN module is not to preserve all information contained in these raw features, but to construct a compact and task-oriented latent representation that retains information useful for distinguishing stocks according to their forward-return characteristics. At the same time, reducing the dimensionality also restricts the number of candidate variables available to GP and therefore helps control the complexity of the subsequent symbolic search.

A. Selection of the Evaluated Range

We evaluate $M \in \{8, 9, 10, 11, 12\}$. This range corresponds to approximately 15.4%–23.1% of the original 52-dimensional feature space, and therefore represents a strong-compression regime. In particular, $M = 10$ corresponds to approximately 19.2% of the original dimensionality. We focus on this compact region because the latent representation is intended to serve as a substantially reduced search space for GP rather than as a reconstruction of the original features.

The range from 8 to 12 also allows us to examine moderate decreases and increases in the latent representation capacity around $M = 10$ while keeping all candidate settings within a comparable low-dimensional regime. If M is excessively small, potentially useful predictive information may be removed during compression. Conversely, an excessively large M weakens the dimensionality-reduction effect, may retain more redundant or noisy information, and increases the number of variables available to GP. Therefore, the evaluated range is designed to examine the trade-off among information preservation, representation compactness, and symbolic-search complexity, rather than assuming that the intrinsic dimensionality of the

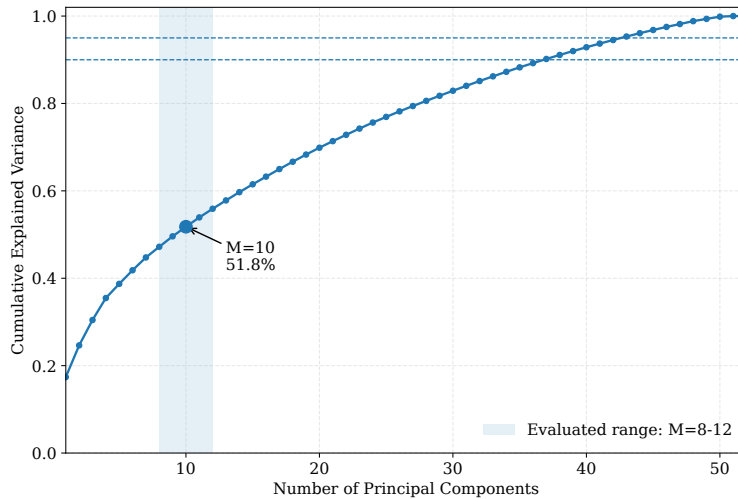


Fig. S4. Cumulative explained variance of PCA on the original 52-dimensional feature space.

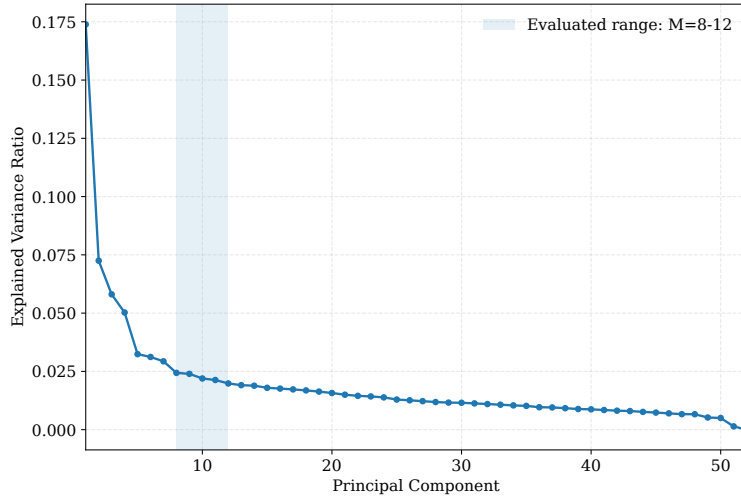


Fig. S5. Explained variance ratio of individual principal components for the original 52-dimensional feature space.

original data is exactly between 8 and 12. The dimensional characteristics of the original feature space and the empirical performance within this compact range are further examined in Figs. S4–S8.

To further understand the structure of the original feature space, we additionally conduct PCA on the training data. As shown in Fig. S4, the original 52-dimensional feature space is not trivially low-rank. Specifically, PCA requires 28 components to explain 80% of the total variance, 32 components for 85%, 37 components for 90%, and 43 components for 95%. The two spectral effective-dimensionality measures are approximately 19.07 and 33.14.

Fig. S5 shows the explained variance contributed by each individual principal component. For the dimensionalities examined in our sensitivity study, the cumulative explained variance under PCA is 47.20%, 49.59%, 51.79%, 53.92%, and 55.90% for $M = 8, 9, 10, 11$, and 12, respectively. In particular, a 10-dimensional linear PCA representation explains only 51.79% of the total variance.

These results indicate that $M = 10$ should not be interpreted as the intrinsic dimensionality of the raw feature space. Instead, they highlight an important distinction between PCA and the proposed CNN representation. PCA is an unsupervised linear transformation designed to preserve the overall variance of the input, whereas our CNN learns a supervised nonlinear representation optimized by the three-class stock-classification objective. Therefore, the CNN does not need to preserve all variance contained in the original features; it is expected to retain information relevant to the downstream return-oriented task while suppressing redundant or less useful information.

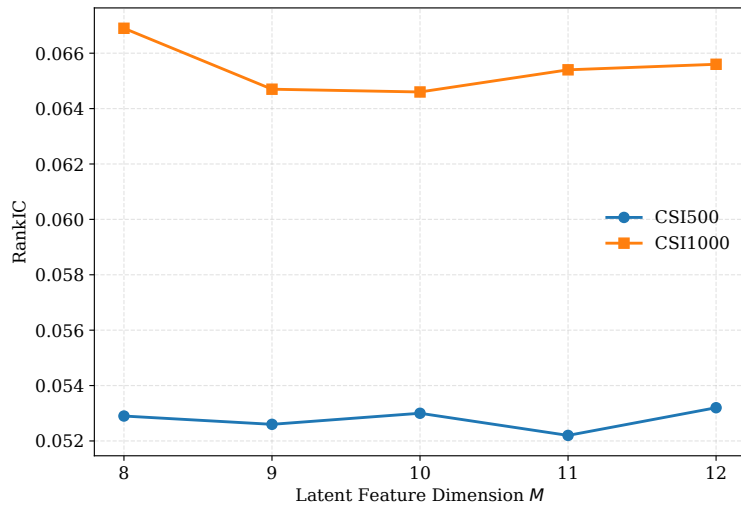


Fig. S6. RankIC under different latent feature dimensions M .

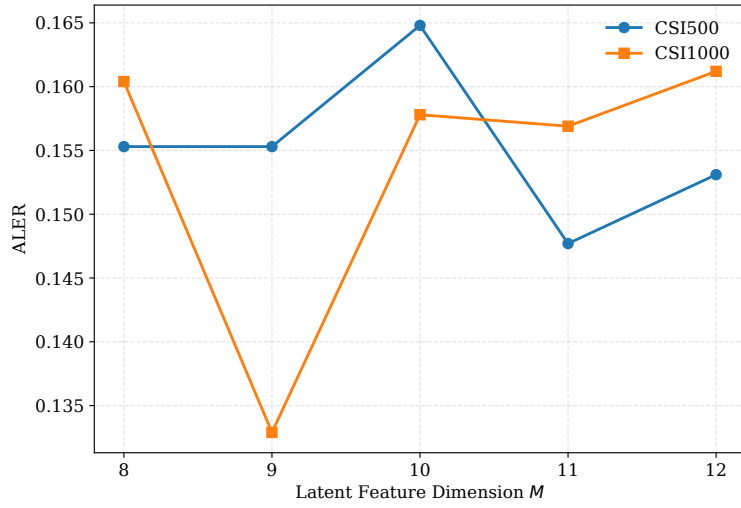


Fig. S7. ALER under different latent feature dimensions M .

In this sense, the fact that CNN-GP can maintain strong predictive performance using only 10 latent variables, even though substantially more PCA components are required to retain most of the raw variance, supports the role of the CNN as a task-oriented feature compressor rather than a conventional variance-preserving dimensionality-reduction method.

B. Sensitivity of RankIC to M

Fig. S6 presents the RankIC performance under different latent feature dimensions M . On CSI500, RankIC values are 0.0529, 0.0526, 0.0530, 0.0522, and 0.0532 for $M = 8, 9, 10, 11$, and 12, respectively. On CSI1000, the corresponding RankIC values are 0.0669, 0.0647, 0.0646, 0.0654, and 0.0656. The RankIC curves therefore show only moderate fluctuations when the latent dimension changes within this compact range.

Importantly, increasing M does not lead to a monotonic improvement in RankIC. For example, CSI500 obtains its highest RankIC at $M = 12$, whereas CSI1000 obtains its highest RankIC at $M = 8$. This suggests that simply retaining a larger latent representation does not necessarily improve cross-sectional ranking ability. Instead, the results indicate that the framework is reasonably robust to moderate variations in M .

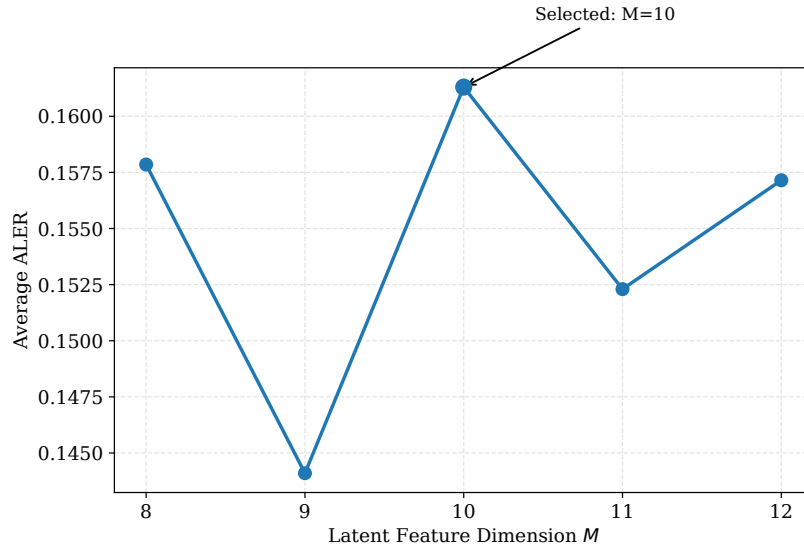


Fig. S8. Average ALER across CSI500 and CSI1000 under different latent feature dimensions M .

C. Sensitivity of ALER to M

Fig. S7 shows the effect of M on ALER for CSI500 and CSI1000. On CSI500, ALER values are 0.1553, 0.1553, 0.1648, 0.1477, and 0.1531 for $M = 8, 9, 10, 11$, and 12 , respectively, with the best result obtained at $M = 10$. On CSI1000, the corresponding values are 0.1604, 0.1329, 0.1578, 0.1569, and 0.1612, where $M = 12$ achieves the highest individual ALER and $M = 10$ remains competitive.

Because ALER is more directly related to realized investment performance, we further consider the average ALER across CSI500 and CSI1000 when selecting the latent dimension. The average ALER values under different latent feature dimensions are shown in Fig. S8.

Among the evaluated settings, $M = 10$ achieves the highest average ALER of 0.16130. Although $M = 10$ is not optimal for every individual dataset–metric combination, it provides the strongest overall investment performance when both stock universes are considered jointly.